

Complex projective varieties and quantum mechanics

Definition:

$$\mathbb{CP}^n = \left\{ (z_0, \dots, z_n) \left| \sum_{j=0}^n |z_j|^2 = 1 \right. \right\}$$

We associate $\mathbb{CP}^n$ to a qudit of dimension n .

The many body Hilbert space we work with is

$$L^2(\mathbb{CP}^n) = \bigoplus_{N \geq 0} h_N$$

which is graded by degree,

$$\deg(p(x)) = N$$

of homogeneous polynomials.

$$|k_0, k_1, \dots, k_n\rangle = \sqrt{\frac{(k_0 + k_1 + \dots + k_n + n)!}{k_0! k_1! \dots k_n! n!}} z_0^{k_0} z_1^{k_1} \dots z_n^{k_n}$$

Forms orthonormal basis w.r.t.

$$\langle k | l \rangle = \int_{\mathbb{CP}^n} \bar{z}^k z^l d\bar{z} dz$$

An ideal (graded) is comprised of all homogeneous polynomials $p(z)$ s.t.

$$(f_1, \dots, f_d)_N = \{p(z) = f_1(z)h_1(z) + \dots + f_d(z)h_d(z) \mid \deg p = N\}$$

We want to realize $(f_1, \dots, f_d)_N$ as ker of Hamiltonian. We need introduce (unnormalised) creation & annihilation operators:

$$a_j^\dagger = z_j, \quad a_j = \partial_j$$

Also write

$$\ell_j^\dagger |k_0, \dots, k_n\rangle = |k_0, \dots, k_j + 1, \dots, k_n\rangle$$

$$\ell_j |k_0, \dots, k_n\rangle = |k_0, \dots, k_j - 1, \dots, k_n\rangle$$

We have

$$\ell_j \propto a_j, \quad \ell_j^\dagger \propto a_j^\dagger$$

The vector subspace $V_{(f_1, \dots, f_d)_N}^\perp$ is defined by the property:

$$q(z) \in V_{(f_1, \dots, f_d)_N}^\perp \iff \langle q(z) | p(z) \rangle = 0 \quad \forall p(z) \in V_{(f_1, \dots, f_d)_N}$$

$$p(z) \in V_{(f_1, \dots, f_d)_N} \implies$$

$$\begin{aligned} p(z) &= \sum_{j=1}^d h_j(z) f_j(z) \\ &= \sum_{j=1}^d a^\dagger(f_j) |h_j\rangle \end{aligned}$$

where

$$a^\dagger(f) = \sum_{k_0, \dots, k_n} f_{k_0 \dots k_n} (a_0^\dagger)^{k_0} (a_1^\dagger)^{k_1} \dots (a_n^\dagger)^{k_n}$$

So

$$\langle q(z) | p(z) \rangle = 0 \implies$$

$$\sum_{j=1}^d \langle q(z) | a^\dagger(f_j) | h_j \rangle = 0 \quad \forall h_j(z)$$

$$\implies \sum_{j=1}^d \langle q | a^\dagger(f_j) | h_j \rangle = 0 \quad \forall |h_j\rangle$$

$$\iff a(f_j) | q \rangle = 0 \quad \forall j = 1, \dots, d$$

$$\iff |q\rangle \in \ker H \quad \text{where}$$

$$H = \sum_{j=1}^d a^\dagger(f_j) a(f_j)$$

Check with example

$$f_0 = z_0 z_1 - z_2^2$$

$$a(f_0) = a_0 a_1 - a_2^2$$

$$[a_0, a_0^\dagger] z_0^k$$

$$[\partial_0, z_0] : \quad \partial_0(z_0 z_0^k) - z_0 \partial_0 z_0^k$$

$$= (k+1)z_0^k - kz_0^k$$

$$= z_0^k$$

$$(a_0 a_1 - a_2^2)(z_0 z_1 - z_2^2) = 1 - 2 \stackrel{?}{=} -1$$

Note:

$$a(f_0)|00\cdots\rangle = 0$$

$$a(f_0)|k_0\cdots k_n\rangle = 0 \quad \text{if } \sum_{j=2}^n k_j = 1$$

$$a(f_0)|112\rangle = |002\rangle - 2|110\rangle$$

$$a(f_1)|11\rangle = |000\rangle$$

$$a(f_0)|002\rangle = -2|000\rangle$$

$$\implies a(f_0)|\psi\rangle = 0 \implies |\psi\rangle = |110\rangle + \frac{1}{2}|002\rangle$$

But this is in $\mathcal{H}$? We realize $V_{(f_1, \dots, f_d)_N}^\perp$ as $\ker H$.

Note $V_{(f_1, \dots, f_d)_N}$.

Is it helpful to have a ground state space given by $V_I^\perp$? Consider, e.g., ideal membership problem:

Suppose we have $\ker H = V_I^\perp$. Is $f \in I$

Basic:

$$H + L_f^\dagger L_f = H'$$

If $f \in I$ then

$$\ker H = \ker(H + H')$$

If $f \notin I$ then

$$\ker(H + H') = ?$$

If $f \notin I$ then

$$\ker(H + H') = ?$$

$$\ker(H + H') = \left\{ |g\rangle \left| \left\langle g \left| \sum_j h_j f_j + h f \right\rangle = 0 \quad \forall h_1, \dots, h_d, h \right. \right\}$$

i.e. $f \notin I$ then

$$I \subset (I, f)$$

and (I, f) is no bigger. So

$$V_{(I, f)} \text{ is smaller.}$$

That is

$$\ker(H + H') \subset \ker(H).$$

Perturbation theory would make value different in limit $p \rightarrow 0$.

Perhaps it is worth trying to build a parent Hamiltonian with

$$\ker(H) = V_I.$$

This should allow finite field.

Strategy: First find parent for single ideal then grow the ideal and add constraints.

Hamiltonian for (f) ?

$$I = (f) = \{h(x)f(x) \mid h(x) \in k[z]\}$$

We start with:

$$\begin{aligned} H &= \mathbb{I} - |f\rangle\langle f| \\ &= \mathbb{I} - L_f^\dagger |0\rangle\langle 0| L_f \end{aligned}$$

Would be good to have unitary version of $L_f^\dagger$.

$$\ker H = \{|g\rangle \mid |\langle g | f \rangle| = 1\}$$

Look now at

$$H = \sum_{h(x)} \mathbb{I} - |hf \times hf|^2 \stackrel{?}{=} \sum_h \mathbb{I} - |u_h f(x)\rangle\langle u_h f(x)|$$

$$(1) = \mathbb{R} \implies H_{(1)} = 0$$

But what is u_f ? Can we use L_f to do this?

For $|f\rangle$:

What $\ker H = \{ |g\rangle \mid |\langle f|g\rangle| = 1 \}$

What equation is satisfied so that

$$M_f|f\rangle = 0$$

$$\Rightarrow L_f - 1$$

$$f(x) = \sum_{j=0}^{\deg(f)} f_j x^j$$

$$|f\rangle = \sum_{j=0}^{\deg(f)} f_j |j\rangle$$

Full basis for $\mathbb{CP}^n$:

$$\mathcal{H} = \text{span} \{ |\alpha_0, \dots, \alpha_n\rangle \mid \alpha_i \in \mathbb{Z}^+; |\alpha\rangle \leftrightarrow z^\alpha \}$$

$$|f\rangle \longmapsto |h_g\rangle ?$$

$$f(x, y) = \sum_{j+k=d} f_{jk} x^j y^k$$

$$h(x, y) = \sum_{l,m} h_{lm} x^l y^m$$

$$f(x, y)h(x, y) = \sum_{\substack{j,k \\ j+k=d}} \sum_{l,m} f_{jk} h_{lm} x^{j+l} y^{k+m}$$

$$|fh\rangle = \sum_{\substack{j,k \\ j+k=d}} \sum_{l,m} f_{jk} h_{lm} |j+l, k+m\rangle$$

If h is homogeneous.

$$|fh\rangle = \sum_{\substack{j,k \\ j+k=d}} \sum_{l,m} f_{jk} h_{lm} |j+l, k+m\rangle$$

If h is homogeneous:

$$|p\rangle \xleftarrow{h \deg(f)} \xrightarrow{h} |hf\rangle \in \mathcal{P}_{\deg(h)+\deg(f)}$$

First look at operator $f \mapsto x^\ell y^m$

$$|j, k\rangle \xrightarrow{x^\ell y^m} |j + \ell, k + m\rangle$$

This is a shift-right operator. $J^\ell |j\rangle = |j + \ell\rangle$

$$J = \sum_{j=0}^{\infty} |j + \ell\rangle \langle j|$$

So this makes Hilbert space interpretation:

$$\mathcal{H}_R \equiv \underbrace{\ell^2(\mathbb{Z}^+) \otimes \ell^2(\mathbb{Z}^+) \otimes \cdots \otimes \ell^2(\mathbb{Z}^+)}_{m+1}_j$$

J is not natural in a Harmonic oscillator setting.

$$\ell^2(\mathbb{Z})$$

would then be appropriate for Laurent polynomials.

So we mirror $\mathbb{F}[x]$ case:

We want to create superposition of all polys in V_I

$$|b\rangle \mapsto \sum_{h \in R_d} |hf\rangle \quad ; \quad R = \bigoplus_{d=0}^{\infty} R_d$$

Is arbitrary enough?

$$|f\rangle \mapsto |f + h\rangle \Rightarrow \infty$$

Is addition unitary?

$$|f\rangle \mapsto |f + h\rangle \Rightarrow \infty$$

But we could start with forming $|k\rangle$.

Thus

$$R = \sum_{\alpha_0, \dots, \alpha_n} |\alpha_0, \alpha_1, \dots, \alpha_n\rangle$$

Assume this is easy to create (how?)

Remember! $|0 \cdots 0\rangle$ stands for “1” etc.

$$1 \mapsto z_0^{\alpha_0} z_1^{\alpha_1} \cdots z_n^{\alpha_n}$$

$$|0 \cdots 0\rangle \mapsto L_{z_0}^{\alpha_0} \cdots L_{z_n}^{\alpha_n} |0 \cdots 0\rangle = |\alpha_0, \alpha_1, \dots, \alpha_n\rangle$$

$\therefore$ we could proceed by applying:

$$\text{CG} = \sum_{j=0}^{\infty} J^j \otimes |j\rangle\langle j|$$

To form

$$\begin{aligned} l(z^\alpha) &= \sum_{\beta} |\alpha + \beta\rangle \\ &= \sum_{\beta} J^\beta |\alpha\rangle \end{aligned}$$

$$l(z_1^{\alpha_1} z_2^{\alpha_2}) = \sum |\alpha_1 + \beta_1\rangle \oplus |\alpha_2 + \beta_2\rangle$$

Eg. $\alpha_1 = (1, 0)$ $\alpha_2 = (0, 1)$

We need to form superpositions:

$$|0\rangle \mapsto \frac{1}{\sqrt{2}} (|\alpha_1\rangle + |\alpha_2\rangle)$$

Actually: this is ok for monomial ideals:

$$(z_1^{\alpha_1}, z_2^{\alpha_2}) = \{p(z) \mid p(z) = h_1(z)z^{\alpha_1} + h_2(z)z^{\alpha_2}\}$$

$$|p(z)\rangle \in \text{span} \{|\alpha_1 + \beta_1\rangle, |\alpha_2 + \beta_2\rangle \mid \beta_1, \beta_2 \in (\mathbb{Z}^+)^{n+1}\}$$

$$(z^{\alpha_1}, z^{\alpha_2}) = \{p(z) \mid p(z) = h_1(z)z^{\alpha_1} + h_2(z)z^{\alpha_2}\}$$

Example.

$$|(z_1, z_2)\rangle \in \text{span} \{|1 + \beta_1^0, \beta_2^0\rangle, |\beta_1^0, 1 + \beta_2^0\rangle\}$$

$$\in \mathcal{H}_N \cap \{|0, 0\rangle\}^\perp$$

The space $|f \cdot h\rangle$ is large enough to characterise

$$|f \cdot h\rangle \in \text{span} \left\{ J^\beta \left(\sum_{\alpha} f_{\alpha} |\alpha\rangle \right) \mid \beta \in (\mathbb{Z}^+)^{n+1} \right\}$$

So, really, we want to start with

$$H = \mathbb{K} - \sum_{j=1}^d |f_j\rangle \langle f_j|$$

$\Leftarrow$ AND quantifier (!)

No, actually, OR quantifier

$$H|f_j\rangle = 0$$

Then sum over all shifts:

$$H \mapsto \sum_{\alpha} J^{\alpha} H (J^{\alpha})^{\dagger}$$

$$H \sum_j c_j |f_j\rangle =$$

$$\sum_j c_j H |f_j\rangle = 0$$

(Can drop $\mathbb{K}$; No, normalisation not important.